FA

FIRE ALARM

GOLDEN STATE INTEGRATED SYSTEMS | LICENSE #1150625

FIRE ALARM SYSTEM DESIGN REFERENCE GUIDE

A Practical Step-by-Step Field Manual

GOVERNING STANDARDS

- NFPA 101 (2021 Ed.) — Life Safety Code
- NFPA 72 (2022 Ed.) — National Fire Alarm & Signaling Code

2024 REFERENCE EDITION

For designer use only. Verify all requirements against AHJ and current adopted code edition.

TABLE OF CONTENTS

HOW TO USE THIS GUIDE **3**

STEP 1 Determine Occupancy Type **4**

STEP 2 New vs. Existing Building **5**

STEP 3 Is a Fire Alarm System Required? **6**

STEP 4 Means of Egress & Evacuation Strategy **7**

STEP 5 Alarm & Notification Type **8**

STEP 6 Technical Design — NFPA 72 .... **9**

STEP 7 Power, Battery & Survivability **11**

STEP 8 Documentation & Code Compliance **12**

HOSPITAL DESIGN CASE STUDY (NFPA 101 Ch. 18) **13**

Occupancy Requirement Matrix **15**

Device Spacing Quick Reference **17**

Submittal Package Checklist **18**

Battery Calculation Worksheet **20**

Glossary of Key Terms **21**

HOW TO USE THIS GUIDE

This guide walks you through every phase of fire alarm system design in the sequence a designer must follow — starting with NFPA 101 (life safety intent) and moving to NFPA 72 (technical implementation). Each step identifies governing code sections, quotes actual code language, and explains exactly what the designer must do in plain terms.

Core Design Philosophy:

- **NFPA 101** tells you IF and WHY a fire alarm system is needed.
- **NFPA 72** tells you HOW to design, install, and document it.
- Your **AHJ** (Authority Having Jurisdiction) may adopt local amendments — always confirm edition and any local changes.
- Fire alarm design is **not** a mechanical code-copying exercise. It requires professional judgment applied to the specific occupancy and risk profile.

PRO TIP Use the Occupancy Matrix (pg. 15) as your first filter. Code language is quoted throughout so you can design without constantly flipping between codebooks. Print this guide and keep it at your workstation.

1 DETERMINE OCCUPANCY TYPE

NFPA 101 Chapters 11–43

The occupancy classification is the most critical starting point. It determines which chapter of NFPA 101 governs the project and sets every downstream life safety requirement — including whether a fire alarm system is required at all.

NFPA 101 §6.1.2

"The occupancy of a building or structure, or portion thereof, shall be classified in accordance with the definitions of occupancies in Chapter 6."

OCCUPANCY CLASSIFICATION REFERENCE

Occupancy Type	New Ch.	Existing Ch.	Typical Examples
Assembly	12	13	Theaters, Arenas, Restaurants (>50 occ.)
Educational	14	15	K-12 Schools, Day Care Centers
Healthcare	18	19	Hospitals, Nursing Homes, ICUs
Ambulatory Care	20	21	Outpatient Surgery, Dialysis Centers
Detention / Correctional	22	23	Jails, Prisons, Detention Facilities
Residential (1&2 Family)	24	24	Single / Two-Family Dwellings
Hotels / Dormitories	28	29	Hotels, Motels, College Dorms
Apartments	30	31	Multi-unit Residential Buildings
Residential Board/Care	32	33	Group Homes, Assisted Living
Mercantile	36	37	Retail Stores, Shopping Malls
Business	38	39	Offices, Banks, Medical Clinics
Industrial	40	40	Factories, Manufacturing Plants
Storage	42	42	Warehouses, Parking Garages
Special Structures	11	11	Towers, Underground Buildings, Tents

Designer Checklist — Step 1:

- Obtain project drawings and confirm the **primary use** of every space.
- Identify any **mixed occupancies** — determine separated vs. non-separated per NFPA 101 §6.1.14.
- Flag the governing NFPA 101 chapter(s) on the design cover sheet.
- Confirm which **edition** the AHJ has adopted — do not assume the current edition.

2 NEW vs. EXISTING BUILDING

NFPA 101 §4.6 + Occupancy Chapter

NFPA 101 applies different requirements depending on whether the building is new construction or an existing building. Each occupancy chapter is split into new and existing sections with potentially different thresholds for when a fire alarm system is required.

NFPA 101 §3.3.150.1 NEW	*"Construction, materials, or designs that do not comply with the requirements that applied at the time the original certificate of occupancy was issued."*
NFPA 101 §3.3.150.2 EXISTING	*"Buildings or conditions that were legally in existence prior to the effective date of this Code."*

Condition	Classification	Design Approach
New ground-up construction	NEW	Apply new construction chapter requirements in full
Certificate of Occupancy < 1 year	NEW	Apply new construction requirements
Renovation / remodel of existing building	EXISTING	Apply existing chapter; additions may trigger new requirements
Change of occupancy classification	NEW (changed area)	New occupancy requirements apply to altered area
Addition to existing building	NEW (addition only)	Addition meets new; remainder stays existing

CRITICAL A change of occupancy classification ALWAYS triggers a full code compliance review for the new classification — even if the physical structure is unchanged. Coordinate with the AHJ and building department before proceeding.

3 IS A FIRE ALARM SYSTEM REQUIRED?

NFPA 101 §9.6 + Occupancy Chapters

Consult the detection, alarm, and communication section within the occupancy chapter. If the answer is yes, NFPA 101 §9.6 defines what type of system is permitted. Then NFPA 72 governs all technical design.

NFPA 101 §9.6.1

"Where required by another section of this Code, fire alarm systems, fire detection systems, and emergency communications systems and their components shall be designed, installed, tested, and maintained in accordance with the applicable requirements of NFPA 72."

Two-Step Code Path:

- **Step A** — Occupancy chapter (e.g., §36.3.4 for mercantile): Is a system required?
- **Step B** — NFPA 101 §9.6: What TYPE of system? Then NFPA 72 for technical design.

TRIGGERING THRESHOLDS BY OCCUPANCY

Occupancy	Code Section	System Required When...
Assembly – New	§12.3.4.1	Occupant load >300; or with automatic sprinkler system
Assembly – Existing	§13.3.4.1	Occupant load >300; AHJ may modify for existing conditions
Educational – New	§14.3.4.1	ALL new educational occupancies require a fire alarm system
Healthcare – New	§18.3.4.1	ALL new healthcare occupancies require a complete fire alarm system
Hotels – New	§28.3.4.1	ALL new hotels and dormitories require a fire alarm system
Hotels – Existing	§29.3.4.1	4+ stories OR more than 75 guest rooms
Apartments – New	§30.3.4.1	ALL new apartment buildings require a fire alarm system
Apartments – Existing	§31.3.4.1	3+ stories OR 11+ dwelling units
Mercantile – New	§36.3.4.1	>12,000 sq ft gross area OR part of a covered mall
Mercantile – Existing	§37.3.4.1	>12,000 sq ft gross area; AHJ discretion applies
Business – New	§38.3.4.1	>12,000 sq ft gross; OR more than 3 stories above grade
Business – Existing	§39.3.4.1	>12,000 sq ft gross; OR more than 3 stories above grade
Industrial	§40.3.4	Required where evacuation delay or process hazard exists
Storage	§42.3.4	High-hazard contents; AHJ discretion; sprinklers may substitute

NOTE

NFPA 101 §9.6.1.4 permits an automatic sprinkler system to substitute for automatic fire detection in SOME occupancies. This does NOT eliminate manual pull stations or occupant notification requirements. Confirm substitution is permitted in the specific occupancy chapter.

4 MEANS OF EGRESS & EVACUATION STRATEGY

NFPA 101 Chapter 7

NFPA 101 §7.2.1.1 *"Means of egress shall consist of three separate and distinct parts: the exit access, the exit, and the exit discharge."*

Component	Definition	Fire Alarm Design Impact
Exit Access	From any occupied point to the exit (corridors, aisles, rooms)	Audible/visible NACs in all corridors and occupied spaces; zero dead zones permitted
Exit	Enclosed travel to exit discharge (stairs, exit doors)	Audible in all exit enclosures; visible if occupancy requires; min. 15 dB above ambient
Exit Discharge	From exit to public way (lobby, exterior)	Notification confirms alarm to occupants exiting; outdoor sounders may be required

Evacuation Strategy Selection:

- **Total Evacuation** — All floors alarm and evacuate simultaneously. Standard for low-rise occupancies.
- **Selective / Zoned Evacuation** — Alarm zone + adjacent zones evacuate first. High-rise, large assembly.
- **Defend-in-Place** — Occupants shelter in safe areas. Used in healthcare per NFPA 101 §18.7.9.
- **Voice Evacuation (EVAC)** — Required for high-rise (>75 ft) per NFPA 101 §11.8.4.1; see NFPA 72 Ch. 24.

NFPA 101 §9.6.3.6 *"Positive alarm sequence shall not delay the evacuation signal more than 3 minutes and shall not be used where the occupancy chapter requires immediate occupant notification."*

5 ALARM & NOTIFICATION TYPE

NFPA 101 §9.6 + Occupancy Chapters

System Type	What It Includes	NFPA 101 Ref.	Typical Occupancy
Manual Fire Alarm	Manual pull stations + audible/visible NACs; no automatic detection	§9.6.2	Small business, low-risk storage
Automatic Detection Only	Smoke/heat detectors with monitoring; may not include full notification	§9.6.1	Sprinklered mercantile
Complete Automatic System	Manual + automatic detection + full audible/visible NACs throughout	§9.6.1–9.6.3	Hotels, apartments, schools, hospitals
EVAC / Emergency Voice	Intelligible voice notification; zoned amps; two-way firefighter phone	§9.6.3.9 / NFPA 72 Ch.24	High-rise >75 ft, large healthcare
Combination System	Integrated with security, HVAC, elevator recall, door release	§9.6.1.6	Complex or multi-use facilities

NFPA 101 §9.6.3.1

"Occupant notification shall be provided to alert occupants of fire or other emergency where required by another section of this Code, either by audible, tactile, or visible means or by a combination thereof."

NFPA 72 §18.4.3

"The sound pressure level of audible notification appliances shall be a minimum of 15 dB above the average ambient sound pressure level or 5 dB above the maximum sound level having a duration of at least 60 seconds, whichever is greater, measured 5 ft (1.5 m) above the floor in the area being served."

6 TECHNICAL DESIGN REQUIREMENTS

NFPA 72 Chapters 10–24

With NFPA 101 requirements established, design the system per NFPA 72. This covers initiating device placement, notification appliance circuits, circuit design, and performance calculations.

6A — INITIATING DEVICES

NFPA 72 Chapter 17

Smoke Detector Placement (NFPA 72 §17.7):

NFPA 72 §17.7.3.2.1

"Spot-type smoke detectors shall be located on the ceiling or, if on a sidewall, between the ceiling and 12 in. (300 mm) down from the ceiling to the top of the detector."

- **Flat ceilings:** 30 ft on center max (900 sq ft each) per §17.7.3.1
- **Sloped ceilings:** First row within 36 in. of peak; remaining rows per §17.7.3.6
- **Beams >= 8 in. deep, <= 4 ft OC:** Treat each bay as a separate room per §17.7.3.7
- **HVAC supply within 36 in.:** Relocate detector or shield from direct airflow

Heat Detector Placement (NFPA 72 §17.6):

NFPA 72 §17.6.3.1.1

"The spacing of heat detectors shall be based on the characteristics of the detectors as listed, the mounting height, and the construction of the space in which they are installed."

- **Listed spacing (flat, 10 ft ceiling):** 50 ft on center (2,500 sq ft each)
- **Ceiling height correction:** Reduce per NFPA 72 Table 17.6.3.5.1 for ceilings >10 ft
- **ROR detectors:** Not suitable where normal rapid temperature changes occur (garages, kitchens)

Manual Pull Stations (NFPA 72 §17.14):

NFPA 72 §17.14.8.1

"Manual fire alarm boxes shall be located within 5 ft (1.5 m) of the exit doorway opening."

- Required at each required exit and exit access door
- Mounted 42–48 in. AFF to the handle; max 200 ft travel distance to nearest station
- Must be visible, unobstructed, and red in color

6B — NOTIFICATION APPLIANCE CIRCUITS (NAC)

NFPA 72 Chapter 18

NFPA 72 §18.5.4.4

"Visible notification appliances in a room or corridor shall be of sufficient intensity and properly located so that the appliance is visible throughout the area."

Space Type	Min. Strobe	Placement Rule
Room <= 20 x 20 ft	15 cd	1 device; ceiling or wall 80–96 in. AFF
Room <= 30 x 30 ft	30 cd	1 device centered in room
Room <= 50 x 50 ft	60 cd	1 device centered in room
Corridor <= 20 ft wide	15 cd	1 per 100 ft; within 15 ft of each end
Corridor > 20 ft wide	Per room table	Apply room table using corridor width
Hotel sleeping rooms	110 cd	Wall-mounted; positioned per §18.5.4.3.3
Sleeping w/ pillow alert	177 cd ceiling OR 110 cd wall	See NFPA 72 §18.5.4.3.3

7 POWER, BATTERY & SURVIVABILITY

NFPA 72 Chapters 10 & 12

NFPA 72 §10.6.1 *"All fire alarm systems shall have a primary power supply and a secondary power supply."*

- **Primary Power:** Dedicated AC branch circuit; labeled at breaker; no GFCI/AFCI breakers
- **Secondary Power (Supervised):** 24 hours standby + 5 min. alarm per §10.6.7.2
- **Secondary Power (Non-Supervised):** 60 hours standby + 5 min. alarm per §10.6.7.2.2

NFPA 72 §10.6.7.2 *"The secondary power supply shall have sufficient capacity to operate the system under maximum alarm load for 5 minutes after 24 hours of operation in the normal (supervisory) condition."*

Battery Sizing Formula:

Parameter	Formula	Notes
Standby Capacity	Standby mA / 1000 × 24 hr × 1.2 = Ah required	Sum all device standby currents; 1.2 = safety factor
Alarm Capacity	Alarm mA / 1000 × 0.0833 hr × 1.2 = Ah required	0.0833 hr = 5 min; sum all devices at full alarm
Total Battery Ah	Standby Ah + Alarm Ah = Required Ah	Round up to next standard battery size

Pathway Survivability (NFPA 72 §12.4):

NFPA 72 §12.4.4 *"Pathways shall be designated as Level 0, Level 1, Level 2, or Level 3 based on their ability to continue to operate during a fire condition."*

Level	Requirement	Common Application
Level 0	No survivability required	Low-risk; AHJ discretion only
Level 1	2-hour rated construction OR 2-hr circuit integrity (CI) cable	Most commercial occupancies
Level 2	2-hr CI cable + dedicated raceway OR listed 2-hr FACP	High-rise, healthcare
Level 3	2-hr CI cable in 2-hr rated enclosure throughout entire pathway	Mission-critical, large healthcare

8 DOCUMENTATION, DRAWINGS & CODE COMPLIANCE

NFPA 72 Chapter 7

NFPA 72 §7.3.1

"Approval of the system shall be obtained prior to system installation. If required by the authority having jurisdiction, the documentation shall be submitted to the AHJ for review prior to installation and shall include a complete description of the system, including all equipment, wiring, and connections."

Required Drawing Content (NFPA 72 §7.3.2):

- Floor plans (all floors) with all device locations drawn to scale — min. 1/8 in. = 1 ft
- Equipment schedule: manufacturer, model, UL listing, quantity, mounting height for every device
- Battery calculation worksheet — standby and alarm load totals with 1.2 safety factor applied
- Voltage drop calculations for all NAC and SLC circuits — max 10% allowable drop
- Riser diagram: FACU, all IDCs, SLCs, NACs, supervisory connections, and point counts per circuit
- Point-to-point wiring diagrams: conductor gauge, color, terminal IDs, and home-run path
- Zone schedule: zone number, type, device count, and location description
- Sequence of Operations (SOO) matrix: every input event mapped to its output action
- Interface diagrams: elevator recall, HVAC shutoff, door holders, suppression system monitoring

PRO TIP

Always include a design cover sheet listing the NFPA editions referenced, AHJ contact info, occupancy classification, system type, and design basis statement. It speeds up plan check and demonstrates professional-level design intent.

HOSPITAL DESIGN CASE STUDY

This case study demonstrates how a designer applies NFPA 101 first to define the life safety intent, then NFPA 72 to implement the technical solution. The hospital scenario is one of the most demanding fire alarm design challenges — high risk occupants, limited mobility, and a defend-in-place strategy instead of full evacuation.

NOTE

Fire alarm design is not a mechanical exercise of copying code paragraphs. A compliant and effective design requires the designer to interpret the intent of both codes and apply them to the specific building occupancy and risk profile. Neither NFPA 101 nor NFPA 72 can anticipate every possible building configuration. Both codes establish minimum requirements and performance objectives, leaving room for professional judgment.

PROJECT SCOPE

New Hospital — NFPA 101 Chapter 18

The building includes:

Space	Life Safety Significance
Patient Sleeping Rooms	Highest risk — patients incapable of self-evacuation; primary driver of defend-in-place strategy
Nurse Stations	Primary staff notification point; annunciation must be immediate and location-specific
Patient Care Corridors	Primary evacuation pathway; full audible/visible notification required
Operating Rooms	Hazardous environment; sensitive equipment; special detector and NAC considerations
Diagnostic & Treatment Areas	Variable occupancy; automatic detection required throughout
Mechanical & Electrical Rooms	Hazardous support areas; heat detectors or specialty detectors as appropriate
Exit Stairways	Secondary egress; audible notification required; Level 1+ pathway survivability

PART 1 — NFPA 101 ANALYSIS

Governing: Chapter 18, Chapter 7, Chapter 9

Step 1: Confirm Occupancy — NFPA 101 Chapter 18 (New Healthcare)

NFPA 101 §18.1.1.5

"The healthcare facilities regulated by this chapter shall be those that provide sleeping accommodations for their occupants and are occupied by persons who are mostly incapable of self-preservation because of age, because of physical or mental disability, or because of security measures not under the occupant's control."

NFPA 101 §18.1.1.9

"The requirements of this chapter shall apply based on the assumption that staff is available in all patient-occupied areas to perform certain emergency control functions as required in other paragraphs of this chapter."

NFPA 101 §18.1.1.3.1

"All healthcare facilities shall be designed, constructed, maintained, and operated to minimize the possibility of a fire emergency requiring the evacuation of occupants."

Interpretation — What These Sections Tell the Designer:

Code Finding	Design Implication
Patients are incapable of self-evacuation (§18.1.1.5)	Immediate full building evacuation is NOT the life safety strategy. Design must support staff response and horizontal re
Staff is assumed present in all patient areas (§18.1.1.9)	Alarm annunciation must notify staff first. Staff initiate emergency procedures — not patients.
Design must minimize the need to evacuate (§18.1.1.3.1)	Compartmentation, smoke control, and fire containment are the primary strategies. Alarm system supports these — no

Step 2: Alarm Annunciation — NFPA 101 §18.3.4.3

NFPA 101 §18.3.4.3

"Alarm signals shall be annunciated in accordance with the provisions of this chapter to alert staff so that they can identify the alarm location, initiate emergency procedures, and direct the appropriate response."

- **Staff notification is the primary objective** — not general occupant notification.
- Alarm must allow staff to identify the **exact smoke compartment** of origin.
- General audible alarm to patients is **intentionally limited** to prevent panic.
- Horizontal relocation to adjacent compartment — not stair evacuation — is the expected response.

PART 2 — NFPA 72 IMPLEMENTATION

How the System Is Designed and Built

Once NFPA 101 defines WHERE and WHY alarm functions are required, the designer applies NFPA 72 to implement the system correctly for this occupancy.

NFPA 101 Requirement	NFPA 72 Implementation	Key NFPA 72 Chapter
Staff-only annunciation in patient areas	Private mode notification at nurse stations and FACU; audible NACs in corridors and common areas only; visual indicators	Ch. 23 — Notification
Sound levels appropriate to environment	SPL calculations per ambient noise levels; operating rooms may require lower output; sleeping rooms 110 cd minimum stro	Ch. 18 — NAC
Smoke detector coverage in corridors and patient areas	Spot-type photoelectric detectors in patient corridors at 30 ft OC; duct smoke detectors at HVAC serving patient areas; hea	Ch. 17 — Initiating Devices
Zoning by smoke compartment	Each smoke compartment is a separate alarm zone; FACU annunciates by zone; staff can identify compartment of origin im	Ch. 23 — System Types
Defend-in-place support	Integration with fire/smoke door holders (release on alarm); smoke control system interface; elevator recall per ASME A17.	Ch. 21 — Emergency Control Functions
Manual pull stations at exits	Manual stations within 5 ft of all required exits; 42–48 in. AFF; staff-activated; excluded from patient room corridors where p	Ch. 17 — §17.14
Pathway survivability	Healthcare facilities require Level 1 minimum (2-hr CI cable or rated construction) for all pathway circuits serving patient car	Ch. 12 — Pathways
Power and battery backup	24-hour standby + 5-min. alarm for supervised system; dedicated AC circuit; battery sizing calculation per §10.6.7.2	Ch. 10 — Power Supplies

Hospital Design Summary — Key Takeaways:

- NFPA 101 Chapter 18 establishes a **defend-in-place** life safety model — full evacuation is not the design goal.
- The fire alarm system's primary function is **staff notification by smoke compartment**, not general occupant alarm.
- NFPA 72 Chapter 23 governs the notification mode (private vs. public) — private mode for most patient areas.
- Every smoke compartment boundary must align with the alarm zone layout in the FACU programming.
- Emergency control functions (door holders, HVAC, elevator recall) are critical integration points — document all in the SOO matrix.
- This hospital scenario demonstrates why NFPA 101 analysis must always come before any NFPA 72 technical decisions.

OCCUPANCY REQUIREMENT MATRIX

First-pass filter for any design. **R** = Required **C** = Conditional (see notes) N = Not required by code (AHJ may still require)

Occupancy	Ch.	Manual Pull	Auto Detect	Audible NAC	Visible NAC	EVAC System	Sprinkler Monitor	24 Hr Battery
Assembly New/Exist	12/13	R	C	R	C†	C‡	C	R
Educational New/Exist	14/15	R	R	R	C†	C‡	C	R
Healthcare New/Exist	18/19	R	R	R	R	C‡	R	R
Ambulatory New/Exist	20/21	R	R	R	C†	N	C	R
Hotels/Dorms New/Exist	28/29	R	R	R	R	C‡	C	R
Apartments New/Exist	30/31	R	R	R	C†	C‡	C	R
Res.Board/Care New/Exist	32/33	R	R	R	C†	N	C	R
Mercantile New/Exist	36/37	R	C	R	C†	N	C	R
Business New/Exist	38/39	C	C	R	C†	N	C	R
Industrial All	40	C	C	C	C†	N	C	R
Storage All	42	C	C	C	C†	N	C	R
HIGH-RISE >75 ft	11.8	R	R	R	R	R	R	R

Matrix Notes:

- **C* Conditional:** Required when occupant load, area, or story threshold is exceeded — see occupancy chapter for exact limits.
- **C† Visible NAC:** Required in public/common areas and sleeping rooms per NFPA 72 §18.5.4 and ADA/IBC §1104.
- **C‡ EVAC:** Required for high-rise (>75 ft), large assembly, and healthcare selective evacuation — confirm with AHJ.
- **Healthcare row (highlighted):** All visible NAC and sprinkler monitor columns are Required — stricter than most occupancies.
- **High-Rise row (highlighted):** All 8 columns are Required. Two-way firefighter phone always required per §11.8.4.1.

DEVICE SPACING QUICK REFERENCE

Device Type	Max Spacing	Ceiling Ht.	Wall Mount?	NFPA 72	Key Exceptions / Notes
Smoke Detector – Spot	30 ft OC 900 sq ft	<=10 ft	Yes – top 12 in.	§17.7.3.1	Reduce near HVAC; beams >8 in. divide bays
Smoke Detector – Linear	Per listing; max 60 ft beam	Varies	No	§17.7.4	Large open areas; mount 4–18 in. from peak
Heat Detector – Spot	50 ft OC 2,500 sq ft	<=10 ft	No	§17.6.3.1	Reduce per Table 17.6.3.5.1 for high ceilings
Heat Detector – Linear	Per listing	Varies	No	§17.6.4	Cold storage, large warehouses
Manual Pull Station	200 ft travel max	N/A	42–48 in. AFF	§17.14.8	Within 5 ft of exit door; red color required
Horn/Strobe – Wall	Per Table 18.5.4.3.1	N/A	80–96 in. AFF	§18.5.4	Corridors: 1 per 100 ft, within 15 ft of ends
Strobe – Ceiling	Per Table 18.5.4.3.2	N/A	Yes	§18.5.4	Synchronize if multiple in same field of view
Speaker – Ceiling	Per acoustical calculation	N/A	Yes	§24.4	Min. 10 dB above ambient; 2 kHz test tone
Duct Smoke Detector	1 per HVAC >2,000 CFM	N/A	In duct only	§17.7.5	Also at return air; HVAC shutoff relay required
CO Detector	Per NFPA 720 / local AHJ	Ceiling pref.	Yes	NFPA 720	Required near sleeping areas by most AHJs

CRITICAL

All spacings are MAXIMUMS for ideal listed conditions. High ceilings, beams, irregular geometry, obstructions, and special environments (freezers, clean rooms, high-airflow) require tighter spacing or specially listed equipment. Always verify against the manufacturer's installation instructions AND UL listing.

SUBMITTAL PACKAGE CHECKLIST

A complete fire alarm submittal package includes all items below. Confirm with the AHJ — some jurisdictions require digital submission or additional forms.

ADMINISTRATIVE DOCUMENTS	DESCRIPTION / REQUIRED CONTENT
■ Permit Application	AHJ form with project address, occupancy, scope of work, contractor license number
■ Design Basis Statement	Narrative: occupancy type, system type, NFPA editions, evacuation strategy, AHJ coordination
■ Contractor License Certificate	C-7, C-28, or equivalent; NICET certification if required by AHJ
■ Insurance Certificate	GL and workers comp; name AHJ/owner as additional insured as required

TECHNICAL DRAWINGS (For Permit)	DESCRIPTION / REQUIRED CONTENT
■ Floor Plans — All Floors	All device locations to scale (min. 1/8 in.=1 ft); legend, North arrow, room labels
■ Riser Diagram	FACU, all circuit types (SLC, IDC, NAC), point counts, pathway class designation
■ Point-to-Point Wiring Diagrams	Conductor gauge, color, terminal IDs, and home-run path for each circuit
■ FACU / Equipment Room Layout	Breaker label, battery location, dedicated circuit ID, grounding detail
■ Sequence of Operations (SOO) Matrix	Every input vs. every output action in tabular format
■ Interface / Relay Diagrams	Elevator recall, HVAC shutoff, door holders, suppression monitoring, BMS

CALCULATIONS & SCHEDULES	DESCRIPTION / REQUIRED CONTENT
■ Battery Calculation Worksheet	Standby + alarm loads; 1.2 safety factor; battery Ah stated (see pg. 20)
■ Voltage Drop Calculations	All NAC circuits; max 10% drop; wire gauge, length, and load documented
■ Device / Equipment Schedule	Manufacturer, model, UL listing, quantity, mounting height for all devices
■ Zone Schedule	Zone number, type, device count, and physical location description

EQUIPMENT DATA	DESCRIPTION / REQUIRED CONTENT
■ Cut Sheets — All Devices	Product data for FACU, detectors, NAC appliances, modules, power supplies
■ UL Listing Certificates	UL 864 (FACU), UL 268 (smoke), UL 521 (heat), UL 1971 (strobe) — all current
■ FACU Programming Summary	Address map, zone assignments, output mappings, alarm routing to supervising station

CLOSEOUT DOCUMENTS (After Installation)	DESCRIPTION / REQUIRED CONTENT
■ Acceptance Test Report	Per NFPA 72 §14.4; completed by qualified tech; AHJ witness may be required
■ As-Built Drawings	All field changes incorporated; final set to owner and AHJ within 30 days
■ System Record of Completion	NFPA 72 §7.8 standard form documenting installed system configuration
■ Owner Training Record	Written evidence of owner/operator instruction on system operation
■ Central Station Certificate	UL 827 certificate and monitoring contract copy for supervised systems

BATTERY CALCULATION WORKSHEET

Complete one worksheet per system. Include in the submittal package. Ref: NFPA 72 §10.6.7.2 — secondary power must support 24 hr standby + 5 min. alarm.

Project Name: ________________________________ Job #: ________________

FACU Make/Model: ________________________________ Date: ________________

Designer: ________________________________ AHJ: ________________

Device / Equipment	Qty	Standby mA ea.	Standby mA Total	Alarm mA ea.	Alarm mA Total
FACU Panel (24 VDC)					
Smoke Detectors					
Heat Detectors					
Monitor Modules					
Control / Relay Modules					
Horn/Strobes (NAC 1)					
Horn/Strobes (NAC 2)					
Remote Annunciator					
Speaker / Amplifier					
Other: ___________					
Other: ___________					
TOTALS			_________ mA		_________ mA

Calculation Steps:

1. Standby Capacity: Total Standby mA / 1000 × 24 hr × 1.2 = _______ Ah required

2. Alarm Capacity: Total Alarm mA / 1000 × 0.0833 hr (5 min) × 1.2 = _______ Ah required

3. Total Required Ah: Standby Ah + Alarm Ah = _______ Ah

4. Battery Selected: _______ Ah (must be ≥ required; select next standard size up)

GLOSSARY OF KEY TERMS

AHJ

Authority Having Jurisdiction — the organization, office, or individual responsible for enforcing code requirements.

CI Cable

Circuit Integrity Cable — fire-rated cable listed to maintain circuit operation for a specified time during fire exposure; required for Level 1+ pathway survivability.

Defend-in-Place

A life safety strategy in which occupants shelter within a protected compartment rather than evacuate. Standard for healthcare occupancies per NFPA 101 Ch. 18.

EVAC / ECS

Emergency Voice/Alarm Communication System — intelligible voice notification per NFPA 72 Ch. 24; required for high-rise and certain healthcare occupancies.

FACU

Fire Alarm Control Unit — the main panel monitoring initiating devices and controlling notification appliances. Must be UL 864 listed.

IDC

Initiating Device Circuit — circuit connecting initiating devices (detectors, pull stations) to the FACU.

NAC

Notification Appliance Circuit — circuit connecting horns, strobes, and speakers to the FACU.

NICET

National Institute for Certification in Engineering Technologies — fire alarm design and inspection certification. Some AHJs require NICET Level II or III on stamped drawings.

Private Mode

Notification intended for specific individuals or a trained staff group — not general occupant notification. Used in healthcare per NFPA 72 §18.3.

ROR

Rate-of-Rise — a heat detector that responds to rapid temperature increases, not a fixed threshold. Not suitable where normal temperature swings occur.

SLC

Signaling Line Circuit — circuit used in addressable systems to communicate with intelligent field devices; allows individual device identification at the FACU.

Smoke Compartment

An area within a building enclosed by smoke barriers; the fundamental unit of horizontal relocation in healthcare facilities.

SOO

Sequence of Operations — a matrix documenting which inputs trigger which output actions in the fire alarm system.

UL 268

UL standard for Smoke Detectors for Fire Alarm Signaling Systems.

UL 864

UL standard for Control Units and Accessories for Fire Alarm Systems — required listing for all FACUs.

Reference document only. All designs must be verified against current adopted editions of NFPA 101 and NFPA 72, applicable local amendments, and AHJ requirements. Golden State Integrated Systems (GSIS) | C7 License #1150625 | Isleton / Stockton, CA